"Redesigning the University Attendance App"

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Course:

User Interface Design (HCI2304)

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Abstract and Introduction

This project aims to redesign the university attendance app to improve usability, simplify navigation, and enhance the overall user experience. The current app has several issues that make tracking attendance difficult for students and faculty. Using user research methods like interviews and surveys, we identified key problems and user needs. Based on these insights, we created new wireframes, visual designs, and an interactive prototype. The goal is to deliver a more efficient, user-friendly attendance solution.

1. User Research

1.1 Target Users

Our target users are university students and instructors.
The focus is on individuals involved classroom attendance processes, especially those who face difficulties with current attendance systems.

1.2 Research Method

We conducted an online survey with 10 participants (7 students and 3 instructors). The survey included multiple-choice and open-ended questions to gather insights about their experience with attendance systems.

1.3 Key Insights

- 80% of participants prefer QR code check-in.
- 60% frequently forget to check in.
- 70% would like to receive notifications when they are late.

1.4 Pain Points

The current system is slow and unreliable.
Users receive no confirmation after checking in.
There are no alerts for missed attendance.

1.5 Personas

Student persona
Name: Sara Al-Qahtani
Age: 21
User Type: University Student
Major: Business Administration
Tech Skills: High uses apps daily for study and organization
Goals:
- Quick and easy attendance.
- Reminders for check-in.
- Access to attendance history.

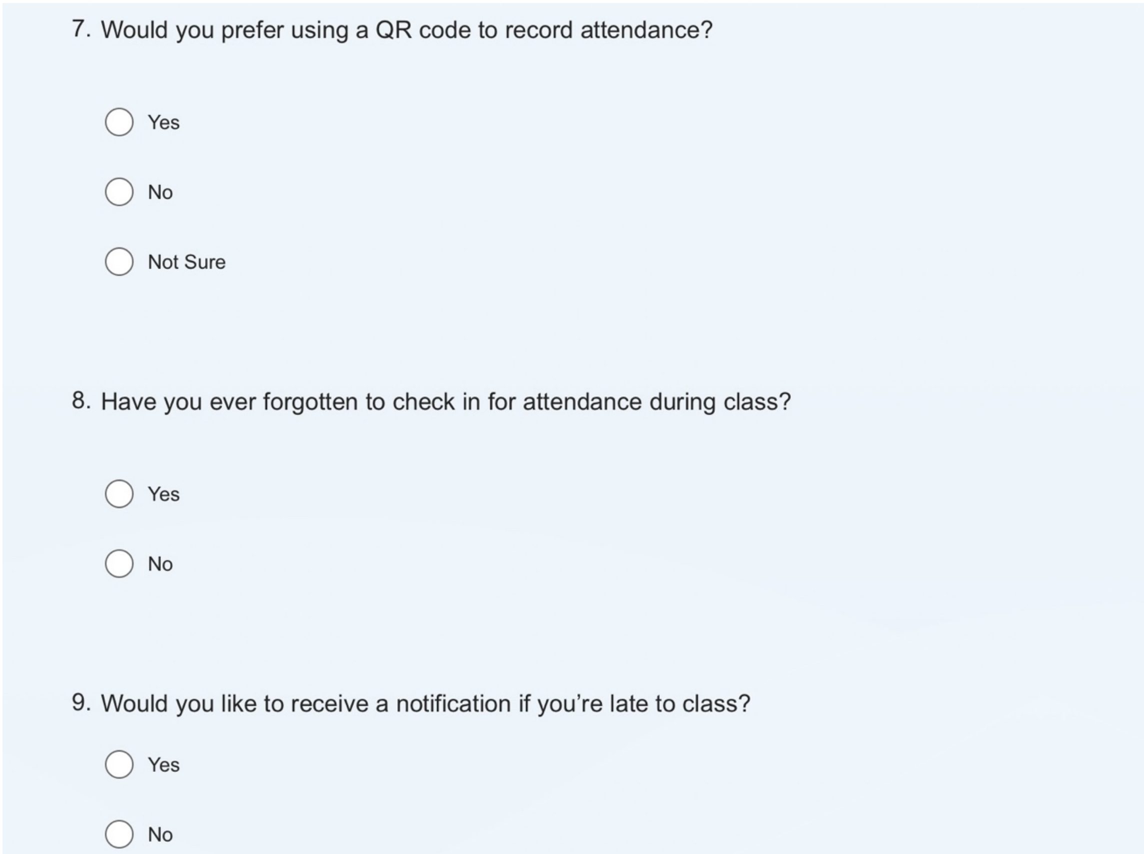
Pain Point:
- Often forgets to check-in.
- No confirmation after check-in.
- App is slow and not user friendly.

Frustrations:
- No alerts for being late.
- Missed check-in affect grades unfairly.

Instructor persona
Name: Dr. Ahmed Al Harbi
Age: 42
Department: Computer Science
Tech Skills: Medium
Goals:
- Accurate attendance tracking.
- Save time during class.
- Get alerts for late or absent.
Pain Points:
- Students claim they checked in, but weren’t recorded.
- Hard to track delays.
- Interface is outdated and unintuitive.
Frustrations:
- Manual double checking of attendance.
- No real-time tracking.

1.6 Research Evidence

Figure1


7. Would you prefer using a QR code to record attendance?

- Yes
- No
- Not Sure

8. Have you ever forgotten to check in for attendance during class?

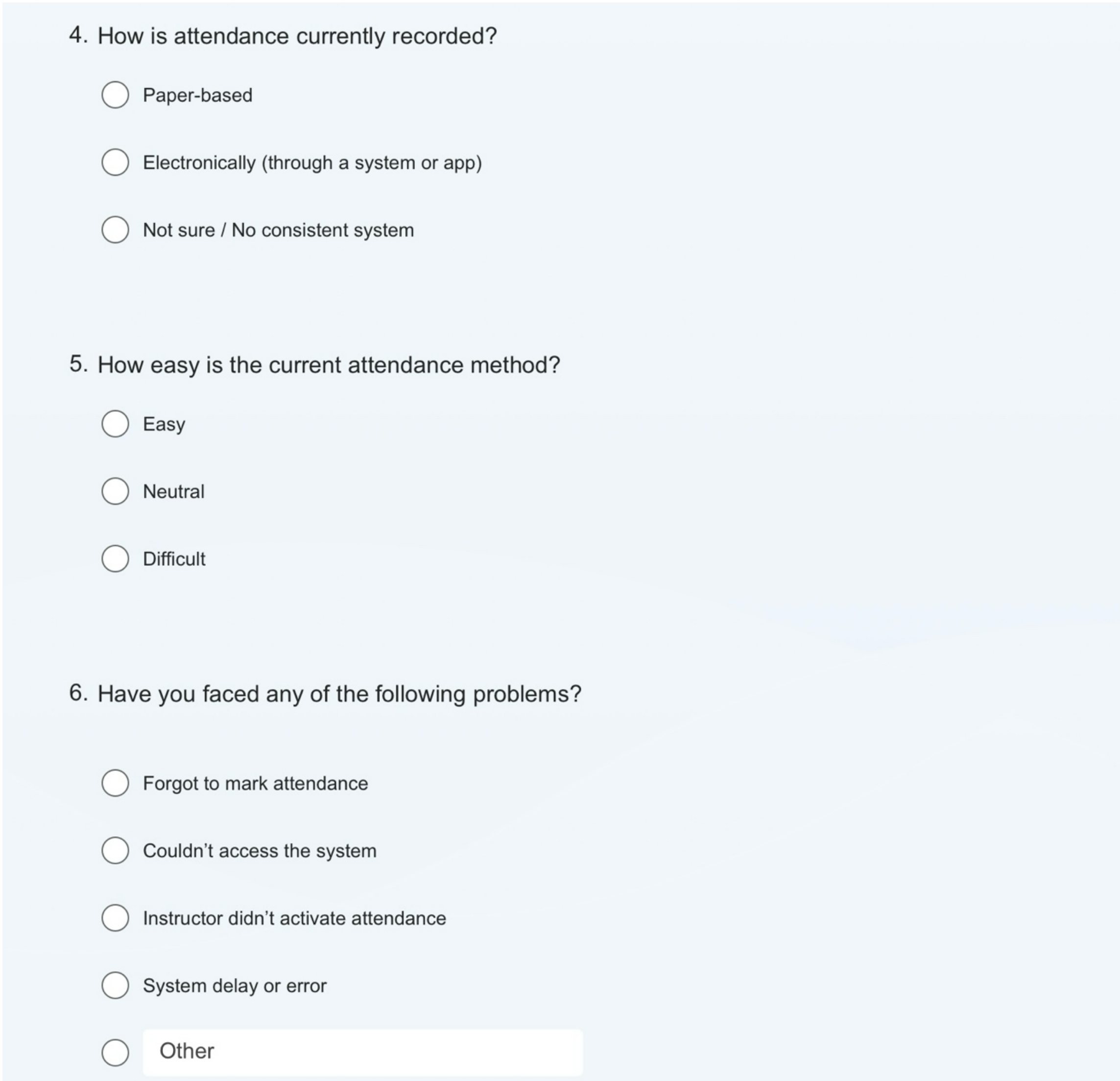
- Yes
- No

9. Would you like to receive a notification if you're late to class?

- Yes
- No

Figure 1: A screenshot of the survey questions shared with participants to gather feedback regarding their experience with the attendance system. These questions specifically target preferences, common issues, and desired features such as QR code check-in and notification reminders.

Figure 2


4. How is attendance currently recorded?

- ◯ Paper-based
- ◯ Electronically (through a system or app)
- ◯ Not sure / No consistent system

5. How easy is the current attendance method?

- ◯ Easy
- ◯ Neutral
- ◯ Difficult

6. Have you faced any of the following problems?

- ◯ Forgot to mark attendance
- ◯ Couldn’t access the system
- ◯ Instructor didn’t activate attendance
- ◯ System delay or error
- ◯ Other

Figure2: A screenshot of the survey questions related to attendance tracking methods. The questions explore how attendance is currently recorded, the ease of the method, and common issues users face.

Summary of Responses:

- One participant mentioned: I forgot to check in twice last week notifications would help.

- Another participant responded: Sometimes I think I checked in, but it didn't go through.

- A third participant said: We use both paper and system methods which makes it confusing.

2. Define Objectives

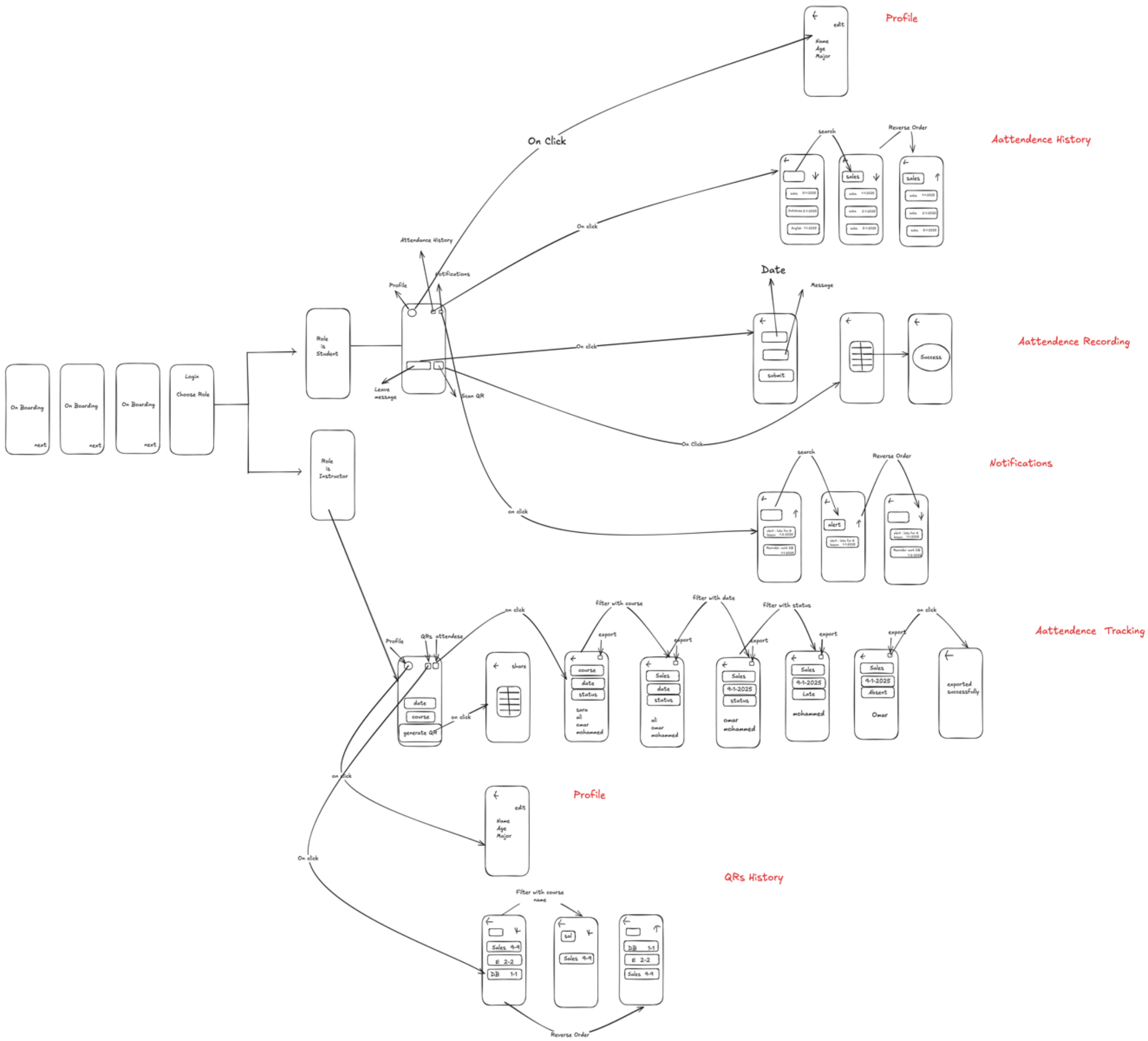
2.1 Project Goals:

- Simplify the attendance process for students and instructors.
- Ensure accurate and real-time attendance records.
- Reduce the number of missed or unrecorded Attendances.
- Notify students immediately if they are late or forgot to check in.
- Improve the user interface to be more intuitive and mobile friendly.

2.2 User Goals:

- **Students**
- Check in quickly via QR code.
- Get alerts for delays or absences.
- Access attendance history easily.
- **Instructors**
- Track attendance in real-time.
- Identify late students instantly.
- Export attendance data easily.

3. Wireframing



4. Visual Design

Figure 1

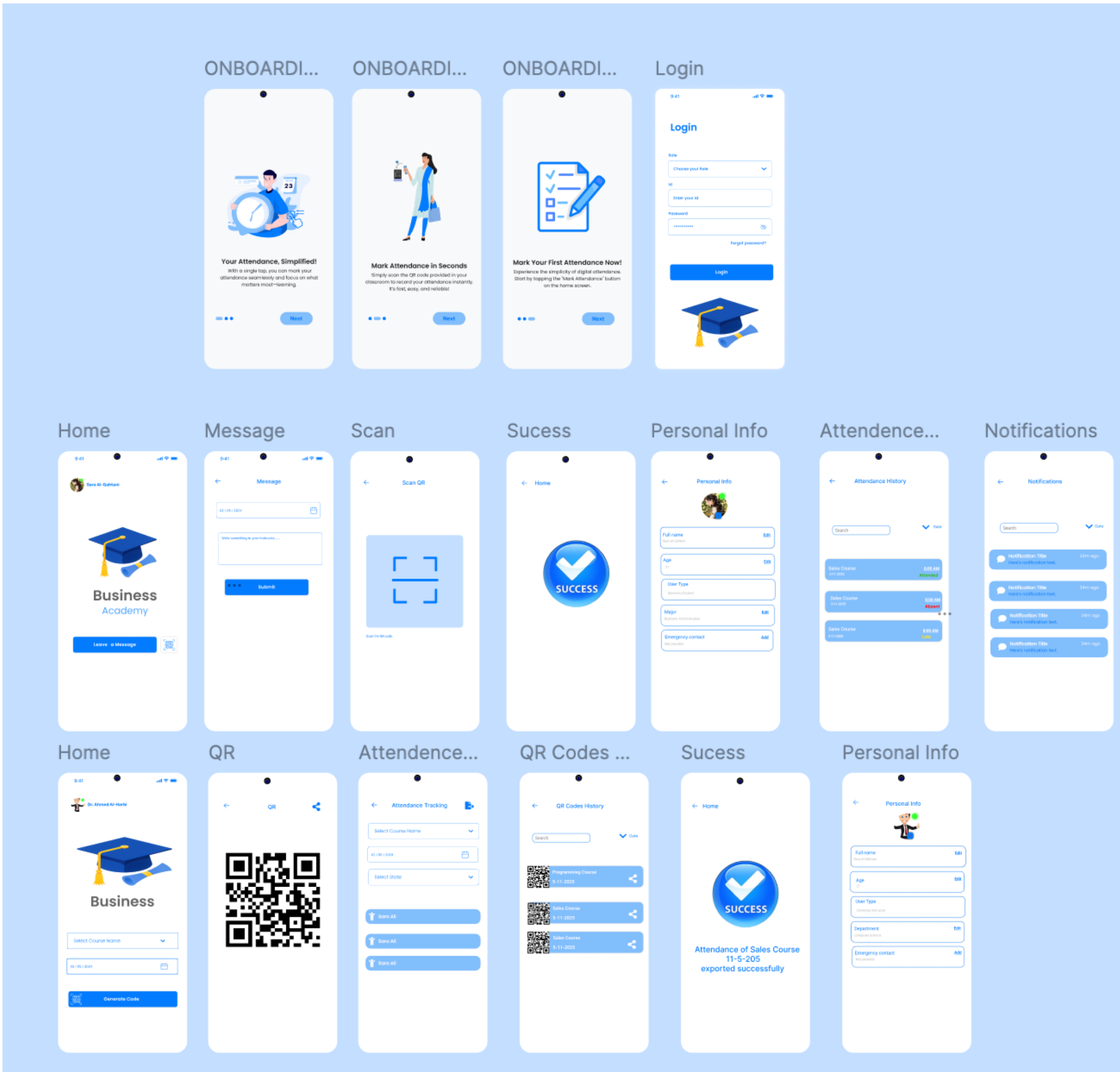


Figure 1: Visual Design Mockups

The interface was designed with a clean and modern look using a consistent blue theme to convey trust, professionalism, and clarity. The onboarding screens introduce the user to the app’s purpose with friendly illustrations. The home screen provides quick access to scan attendance via QR code. The check-in process is simplified with large buttons and confirmation messages, ensuring users like Sara can mark attendance quickly and with confidence. For instructors like Dr. Ahmed, the notifications and history screens provide real time data on student attendance and delays. Overall, the design aims to reduce confusion, enhance usability, and support both student and instructor goals as identified in the research phase.

5. **Prototyping**

After finalizing the visual design, we developed an interactive prototype using Figma to simulate the user experience. The prototype connects the main screens of the application, including the onboarding, login, QR check-in, success confirmation, and attendance history.

This interactive model demonstrates the key user flow: starting from launching the app, scanning a QR code, and receiving confirmation of attendance. The prototype also reflects how students can view their attendance history and how instructors can track attendance notifications.

The prototype helps visualize how the user interacts with the system and allows for early feedback before actual development.

Prototype Link:
https://www.figma.com/design/yBZUY1xFY5YUoe3PJBd4Mr/Attendance-mobile-app?node-id=1-2825&t=rVoPmnQJaPQc9Qpv-1

6. Testing and Iteration

We conducted usability testing with 3 users (3students) to evaluate the prototype and gather feedback. Participants were asked to interact with the main features of the application, such as QR check-in, viewing attendance history, and receiving notifications.

The feedback was overwhelmingly positive. Users reported that the interface was clear, visually appealing, and easy to navigate. No significant usability issues were identified, and participants confirmed that the prototype met their expectations and needs.

As a result, no major changes were required, and the design remained consistent with the original version. This validation reinforced our design decisions and confirmed the overall usability of the product.

7. Development Handoff

7.1 Color Palette

Usage	Hex Code
Primary Blue	#007BFF
Secondary White	#FFFFFF
Accent (Success)	#28C76F
Alert/Red	#FF4D4F
Text Blank	#000000

7.2 Typography

Element	Font/Weight	Size	Example
Titles/Headers	Inter/Bold	24-28px	Login,Success
Subtitles	Inter/Medium	18-20px	Form labels
Body Text	Inter/Regular	14-16px	Descriptions, notifications
Buttons	Inter/Bold	16px	Login, Submit

7.3 Spacing & Layout

Screen Padding: 16-24px

Component Gap: 12-16px

Button Height: 48px

Input Height: 48px

Corner Radius: 8px

7.4 Components

Buttons:

- Primary Button: Blue BG, White Text, 8px Radius
- Disabled Button: Gray BG, Gray Text

Input:

- Full-width, rounded
- Focused: Blue border
- Default: light gray border

Cards:

- Used in notification and QR list
- White background
- Light shadow
- 16px padding

7.5 Iconography & Images

- Outline-style icons
- Profile image: Circular, 48x48px
- Onboarding uses vector-style illustrations

7.6 Screen Flows

1. Onboarding (x3) → Login → Home → Scan QR / Message / Personal Info / Attendance History / Notifications → Success Confirmation

2. QR Code Generation Flow → QR → QR Codes History → Success

7.7 Developer Notes

- Use 8pt grid system for all layouts

- All screens designed at 375x812px

- Ensure accessibility contrast

- Reuse component styles across screens

- Animations should be simple fades/slides

Table Task

Phase	Task	Hind	Hajar	Sham
User Research	Identify target users, use at least 2 research methods, create personas	✓	✓	
Define Objective	Define project goals and user problems			✓
Wireframing	Create low-fidelity wireframes using tools like Balsamiq or Figma		✓	✓
Visual Design	Design UI with suitable colors/fonts, create high-fidelity mockups	✓	✓	✓
Prototyping	Convert design into an interactive prototype	✓	✓	✓
Testing and Iteration	Test with real users and improve based on feedback	✓	✓	
Development Handoff	Prepare design handoff with specs, optionally create a style guide	✓		✓